Universität Stuttgart

Evaluating benchmarks using Bayesian multi-model comparison

Rebecca Kohlhaas

Supervisor: Bernd Flemisch, Sergey Oladyshkin

IWS, University of Stuttgart, Germany

SFB 1313

SimTech

Presence of
reference values?

Evaluation points

QOI

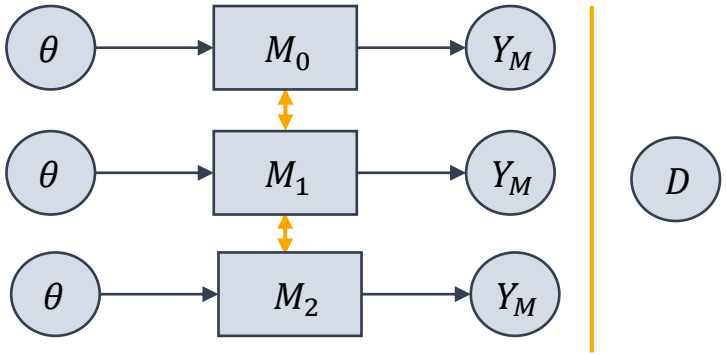
How to evaluate the benchmark?

Grid

Evaluation metric **Bayesian approach to model comparison**

- Intro to Bayesian model comparison
- Application example
- Applicability to (SciML) benchmarks

Uncertainty-aware model comparison



Which model is most plausible given the data D and our prior beliefs?

Model and Input Uncertainties

Experimental Uncertainties and Limitations

Numerical Errors and Uncertainties

Bayesian approach

- Incorporate prior knowledge
- Quantify uncertainty in model selection

Smith (2014)

Bayes' Theorem

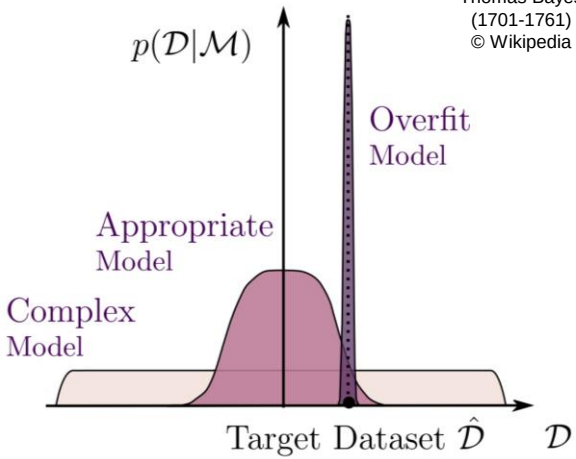
Thomas Bayes
(1701-1761)
© Wikipedia

$$P(\theta|D, M_k) = \frac{P(D|\theta, M_k)\ P(\theta|M_k)}{P(D|M_k)}$$

Marginal likelihood

$$P(D|M_k) = \int_{\Omega_\theta} P(D|\theta, M_k) P(\theta|M_k) d\theta$$

→ Naturally encodes *Occam's razor*



Empirical Bayes / type-II maximum likelihood

$$M_* = argmax_M P(D|M_i)$$

$$P(M_k|D) = \frac{P(D|M_k)\ P(M_k)}{P(D)}$$

Lotfi et al. (2022), Immer. et al. (2021), Schöniger et al. (2014)

Bayesian Model Comparison

Bayesian hypothesis testing

"Model M_k is better than M_l given the observation D"

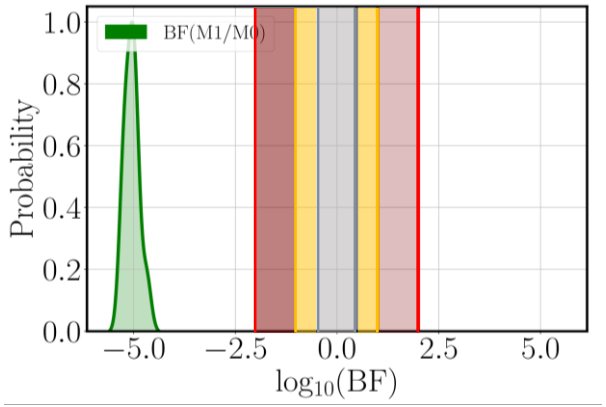
→ Pairwise comparison

Bayes Factor

$$BF(M_k|M_l) = \frac{P(D|M_k)}{P(D|M_l)}$$

Evidence in favor of $\mathbf{M_k}$

- $1 < BF < 3$: "No substantial" evidence
- $BF < 10$: "substantial" evidence
- $BF < 100$: "strong" evidence
- $100 < BF$: "decisive" evidence



Bayesian Model Comparison

Bayesian hypothesis testing

"Model M_k is better than M_l given the observation D"

→ Pairwise comparison

Bayes Factor

$$BF(M_k|M_l) = \frac{P(D|M_k)}{P(D|M_l)}$$

Evidence in favor of $\mathbf{M_k}$

- $1 < BF < 3$: "No substantial" evidence
- $BF < 10$: "substantial" evidence
- $BF < 100$: "strong" evidence
- $100 < BF$: "decisive" evidence

Bayesian model averaging

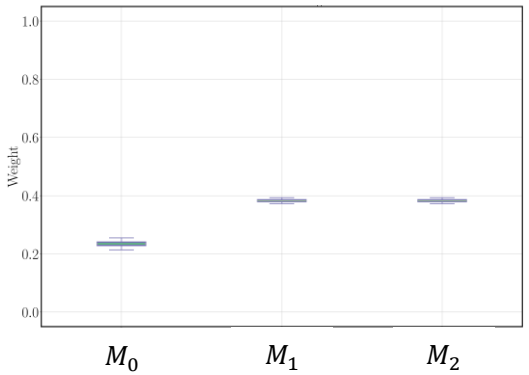
"The optimal model is a combination of the models in the set"

→ Assumes completeness of model set

Model weights

$$P''_{pred}(\theta|Y) = \sum_{k=1}^{N_m} p(\theta|D, M_k) P(M_k|D)$$

$$P(M_k|D) = \frac{P(D|M_k)P(M_k)}{\sum_j P(D|M_j)P(M_j)}$$



Bayesian Model Comparison

Bayesian hypothesis testing

"Model M_k is better than M_l given the observation D"

→ Pairwise comparison

Bayes Factor

$$BF(M_k|M_l) = \frac{P(D|M_k)}{P(D|M_l)}$$

Evidence in favor of $\mathbf{M_k}$

- $1 < BF < 3$: "No substantial" evidence
- $BF < 10$: "substantial" evidence
- $BF < 100$: "strong" evidence
- $100 < BF$: "decisive" evidence

Bayesian model averaging

"The optimal model is a combination of the models in the set"

→ Assumes completeness of model set

Model weights

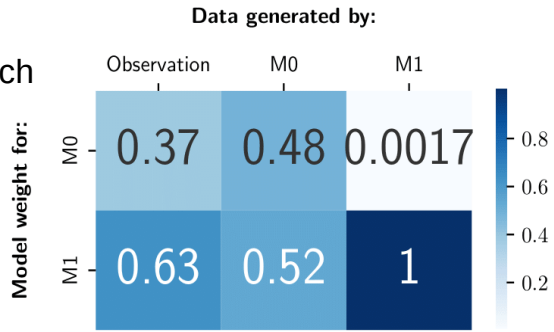
$$P''_{pred}(\theta|Y) = \sum_{k=1}^{N_m} p(\theta|D, M_k) P(M_k|D)$$

$$P(M_k|D) = \frac{P(D|M_k)P(M_k)}{\sum_j P(D|M_j)P(M_j)}$$

Confusion matrix

"Compare the models based on each others results"

$$PW_{lk} = \frac{1}{N_{MC}} \sum_{j=1}^{N_{MC}} p(M_l|M_{k,j})$$



Kass & Raftery (1995), Jeffreys (1998), Schöniger et al. (2015)

Account for surrogate models

Surrogate approximation $\widetilde{M}_k = M_k + E_k$

Correct the BME $P(D|M_k) = P(D|\widetilde{M}_k) \cdot W_{\widetilde{M}_k}$

$$W_{\widetilde{M}_k} = \int_{\Omega_\theta} P\left(M_k \middle| \widetilde{M}_k, \theta\right) P\left(\theta | \widetilde{M}_k, Y\right) d\theta$$

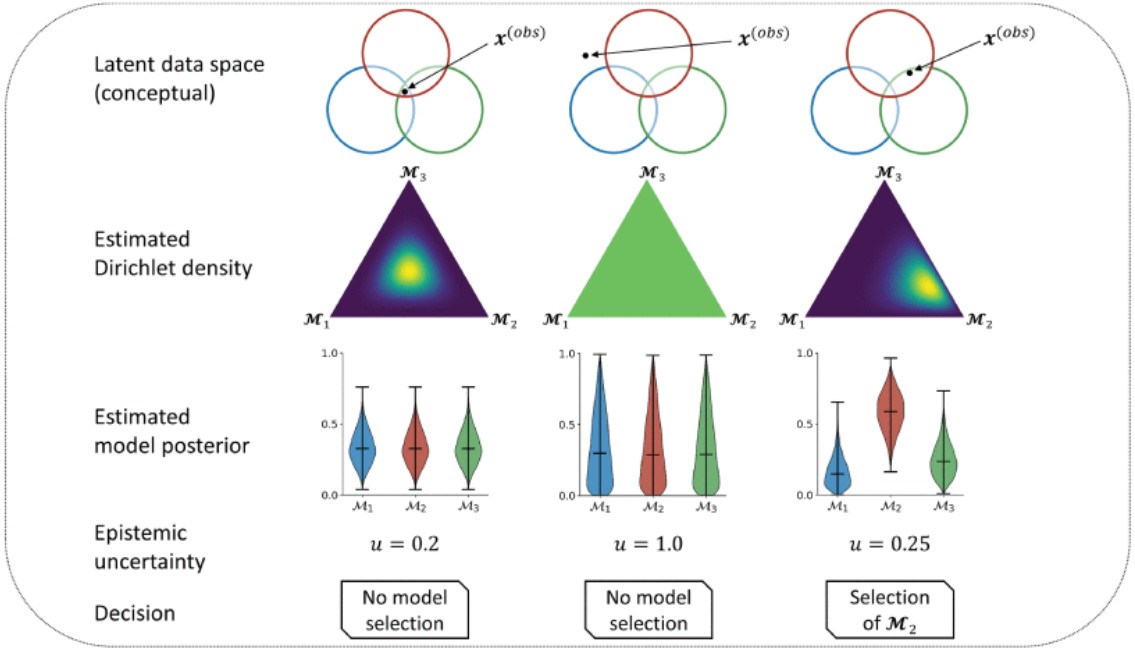
Correct the confusion matrix

$$PW_{lk} = \frac{1}{N_{MC}} \sum_{j=1}^{N_{MC}} p\left(M_l | M_{k,j}\right) = \frac{1}{N_{MC}} \sum_{j=1}^{N_{MC}} p\left(\widetilde{M}_l | \widetilde{M}_{k,j}\right) \cdot W_{\widetilde{M}_k} \cdot W_{\widetilde{M}_l}$$

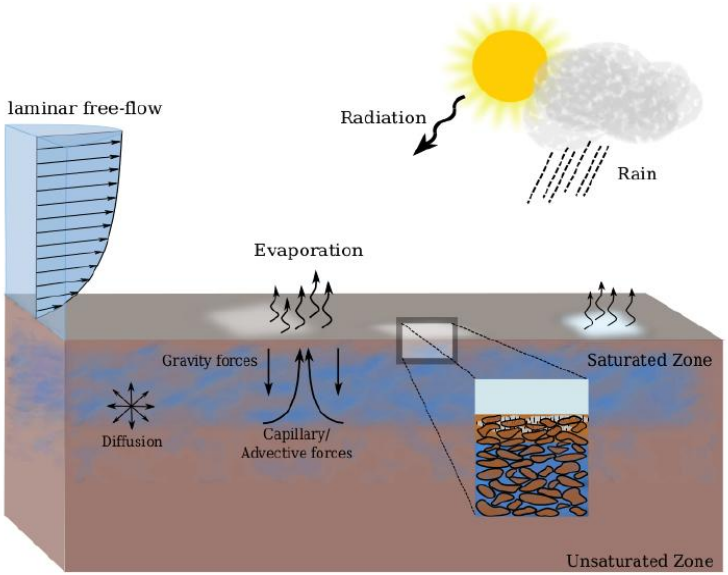
Other methods – Amortized Bayesian Model Comparison

Subjective Logic: Capture epistemic uncertainty using second-order probabilities

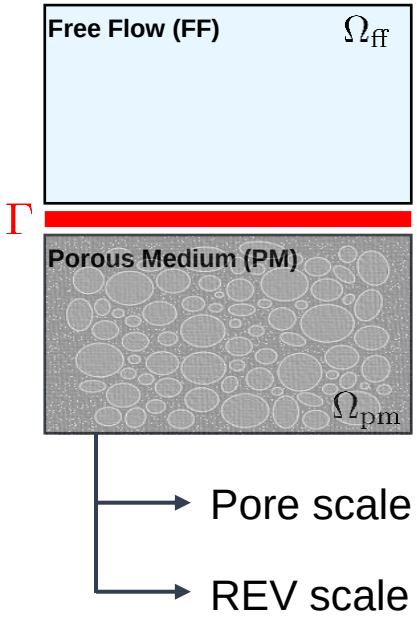
→ Learn properties of a **Dirichlet distribution** via an **Evidence network**



Application: Coupling free flow and porous-medium flow

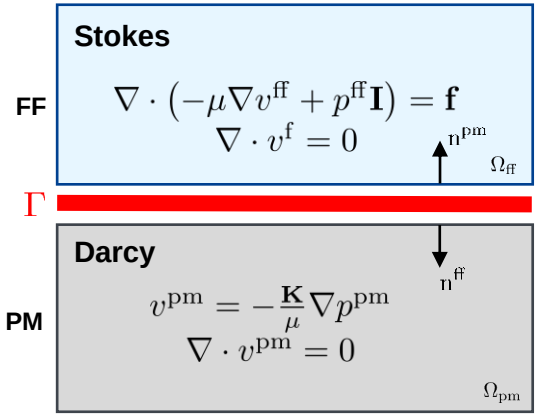


Pavuluri, S., (2014)



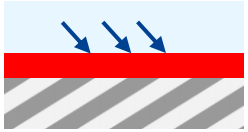
Mohammadi et al. (2023)

Application: Coupling free flow and porous-medium flow



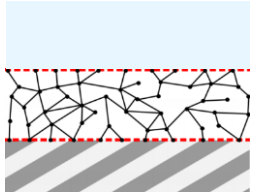
Classical IC

Beavers-Joseph condition[1]



Generalized IC

For arbitrary flows[2]



Pore-network model

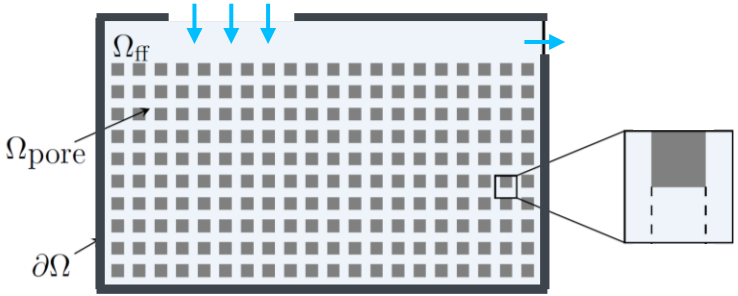
Discrete coupling condition[3]

[1]Beavers & Joseph (1967) [2]Eggenweiler & Rybak (2021) [3]Weishaupt, et al. (2020)

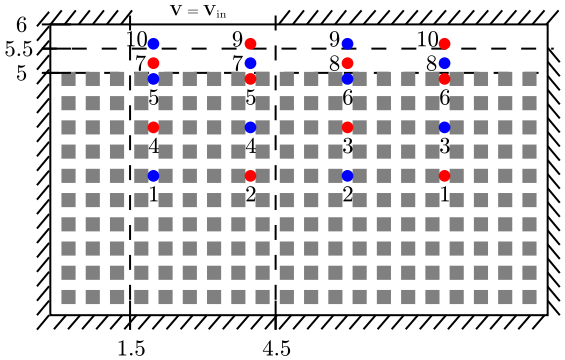
Application: Coupling free flow and porous-medium flow

Stokes Problem

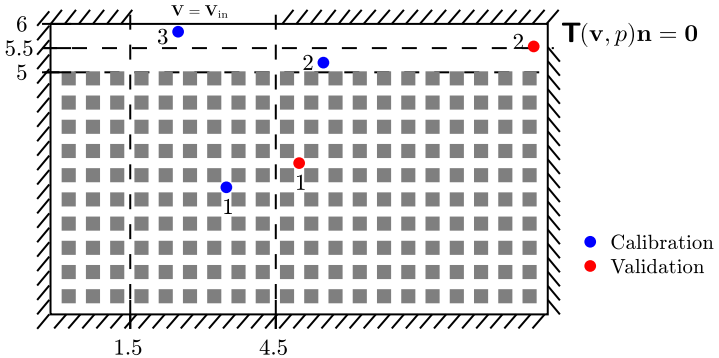
$$\nabla \cdot \mathbf{v} = 0, \quad -\nabla \cdot \mathbf{T}(\mathbf{v}, p) = 0 \quad \text{in } \Omega_{\text{flow}}$$
$$\mathrm{T}(\mathbf{v}, p) = \mu \nabla \mathbf{v} - p\mathbf{I}$$



Velocity



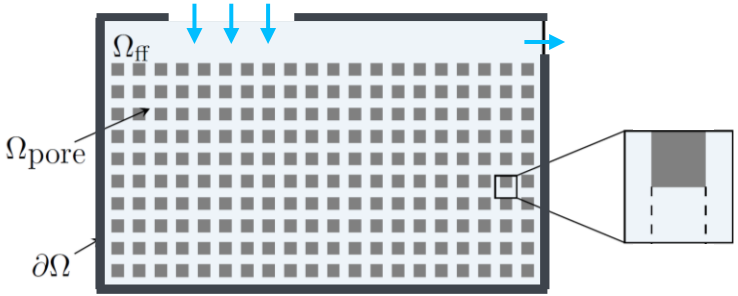
Pressure



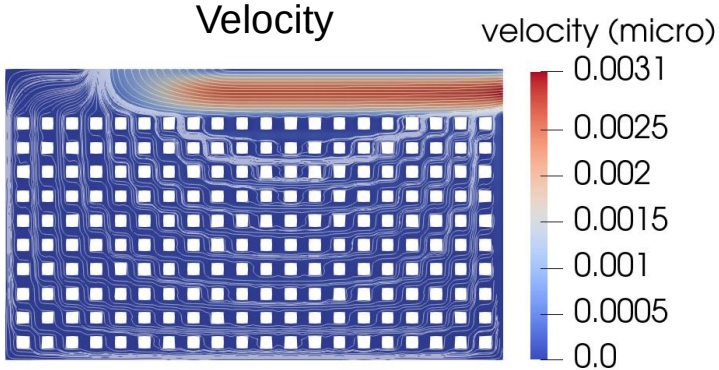
Application: Coupling free flow and porous-medium flow

Stokes Problem

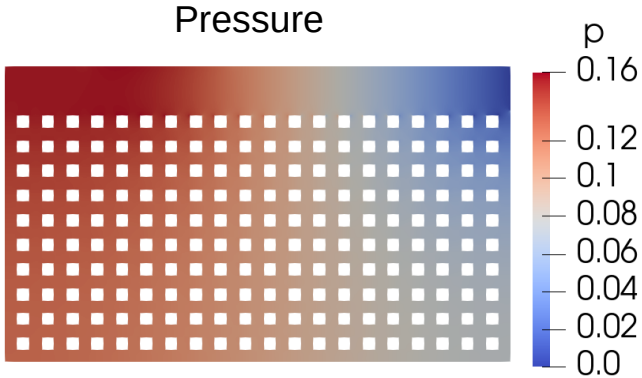
$$\nabla \cdot \mathbf{v} = 0, \quad -\nabla \cdot \mathbf{T}(\mathbf{v}, p) = 0 \quad \text{in } \Omega_{\text{flow}}$$
$$\mathrm{T}(\mathbf{v}, p) = \mu \nabla \mathbf{v} - p\mathbf{I}$$



Velocity



Pressure



Application: Coupling free flow and porous-medium flow

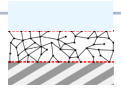
Models → Reference → **Priors** → Sensitivity → Posterior → Bayes Factor

BayesValidRox

Boundary velocity V^{top}	$[5 \cdot 10^{-4}, 1.5 \cdot 10^{-3}]$
Exact interface location γ	$[4.9, 5.1]$
Permeability k	$[10^{-10}, 10^{-8}]$
Beavers-Joseph parameter α_{BJ}	$[0.1, 4]$

Boundary velocity V^{top}	$[5 \cdot 10^{-4}, 1.5 \cdot 10^{-3}]$
Exact interface location γ	$[5.0, 5.1]$
Permeability k	$[10^{-10}, 10^{-8}]$

Boundary velocity V^{top}	$[5 \cdot 10^{-4}, 1.5 \cdot 10^{-3}]$
Total conductance g_{ij}	$[10^{-7}, 10^{-5}]$
Pore-local slip coefficient β_{pore}	$[10^{3}, 10^{5}]$



Measurement Error

Numerical Error

$$f_k = \bar{f} + g_p h_k^{\hat{p}} + \mathcal{O}(h_k^{\hat{p}+1})$$

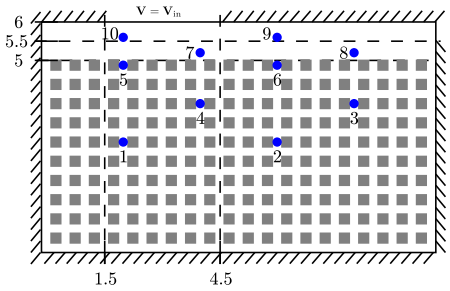
Surrogate Model

Polynomial Chaos Expansion (PCE)

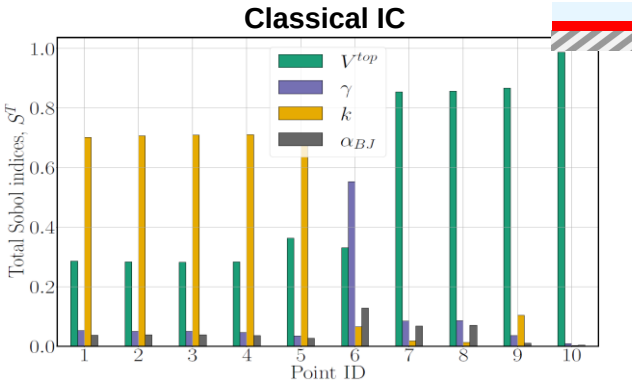
Training: 300 samples

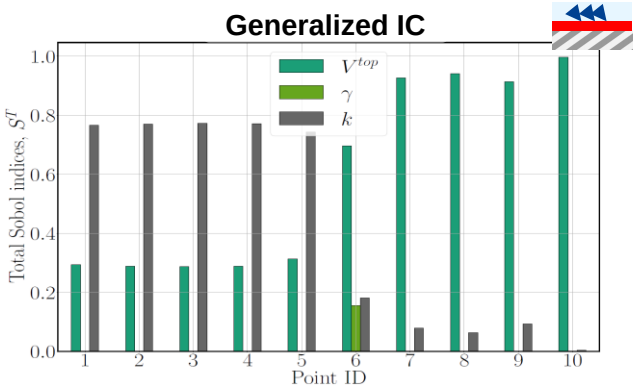
Application: Coupling free flow and porous-medium flow

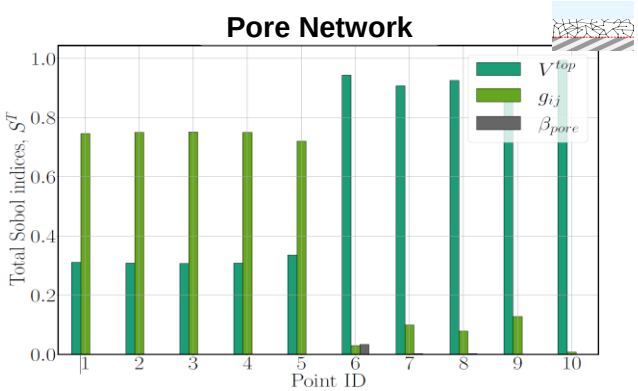
Models → Reference → Priors → Sensitivity → Posterior → Bayes Factor



- Calibration points

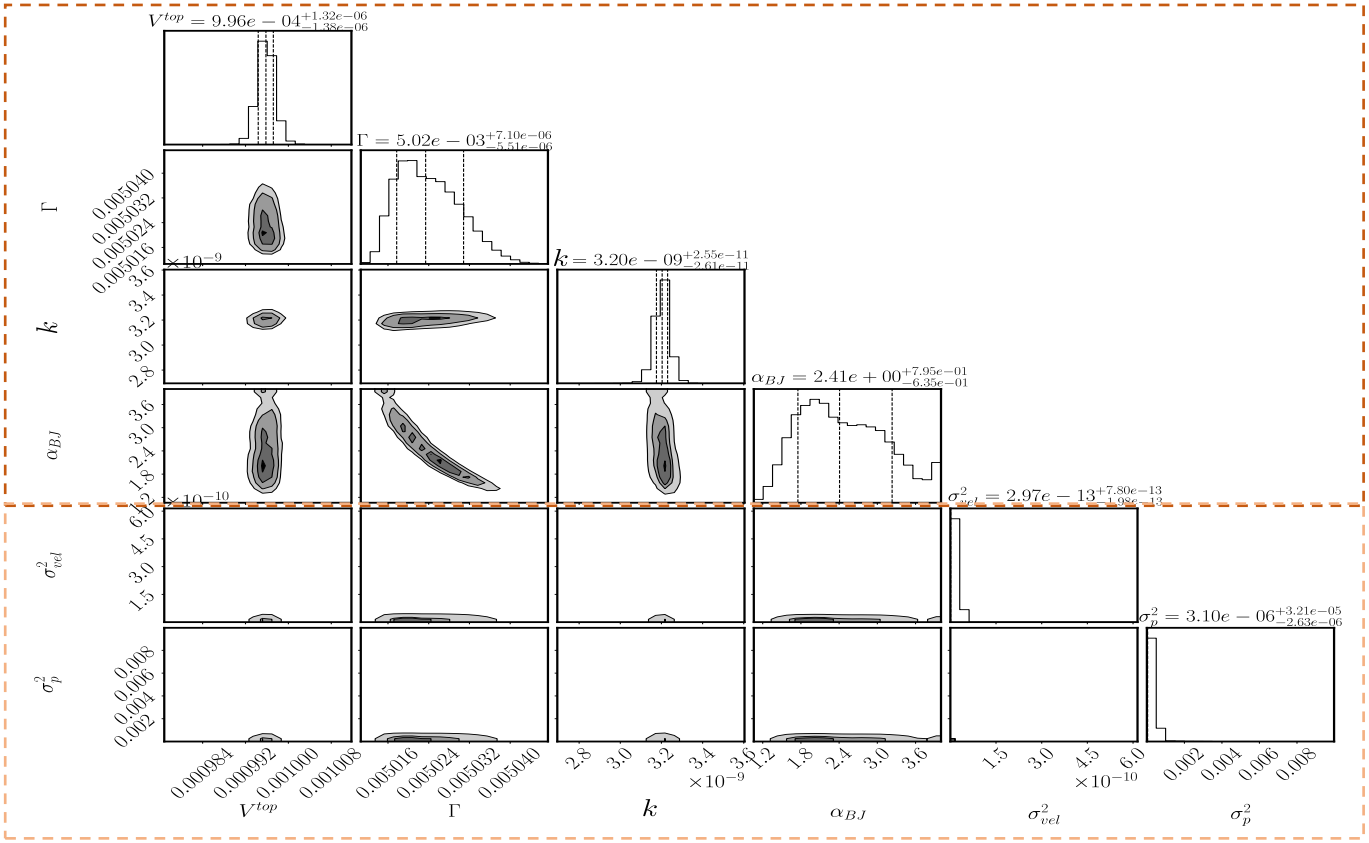






Application: Coupling free flow and porous-medium flow

Models → Reference → Priors → Sensitivity → Posterior → Bayes Factor



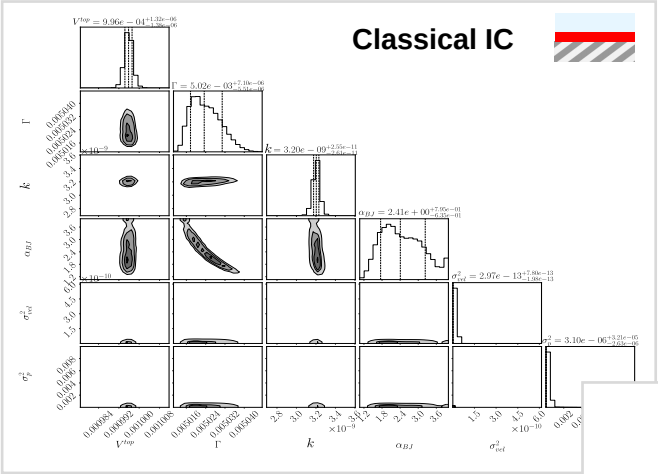
Classical IC

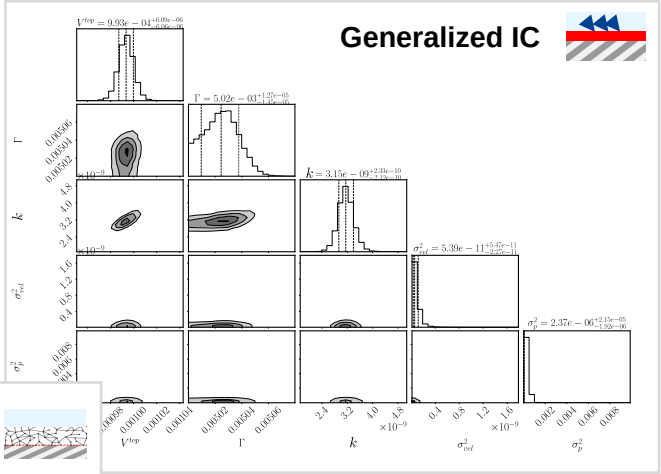
Physical parameters

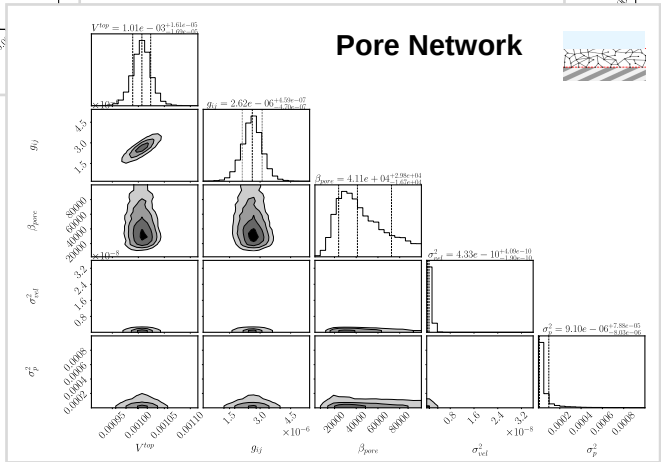
Model discrepancy

Application: Coupling free flow and porous-medium flow

Models → Reference → Priors → Sensitivity → Posterior → Bayes Factor

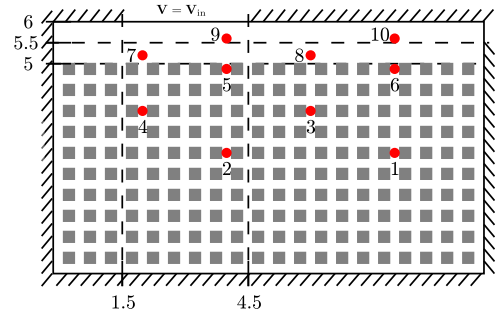






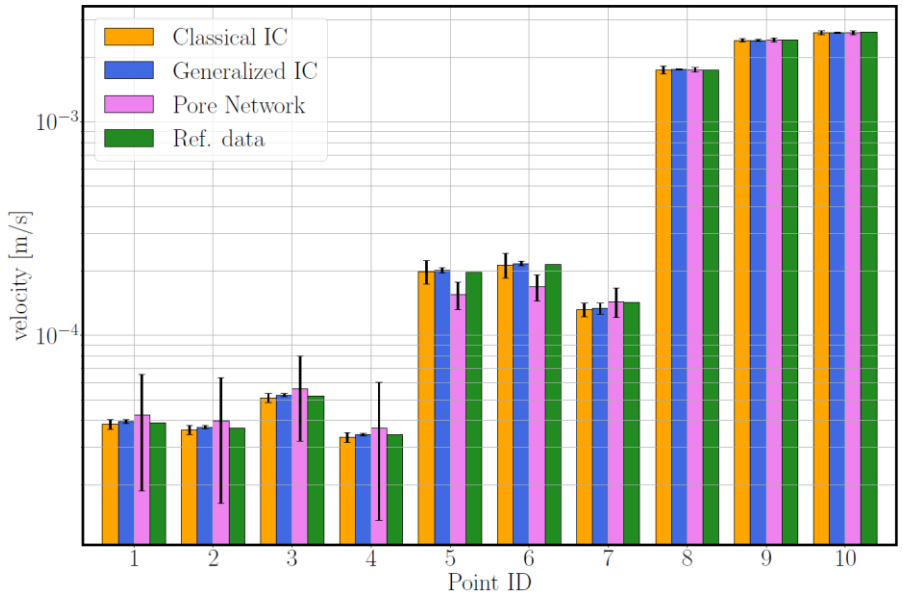
Application: Coupling free flow and porous-medium flow

Models → Reference → Priors → Sensitivity → **Posterior** → Bayes Factor

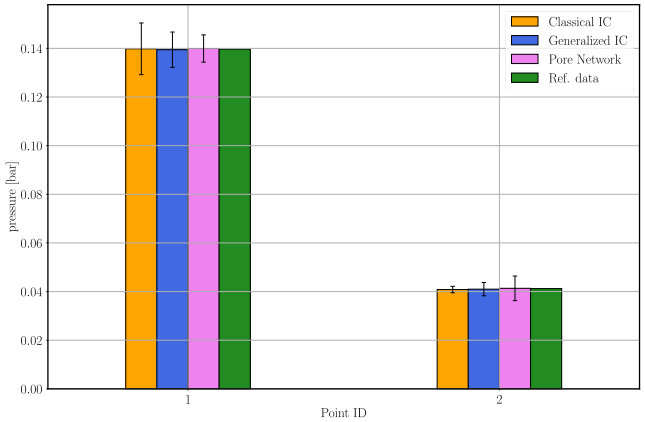


● Validation points

Velocity

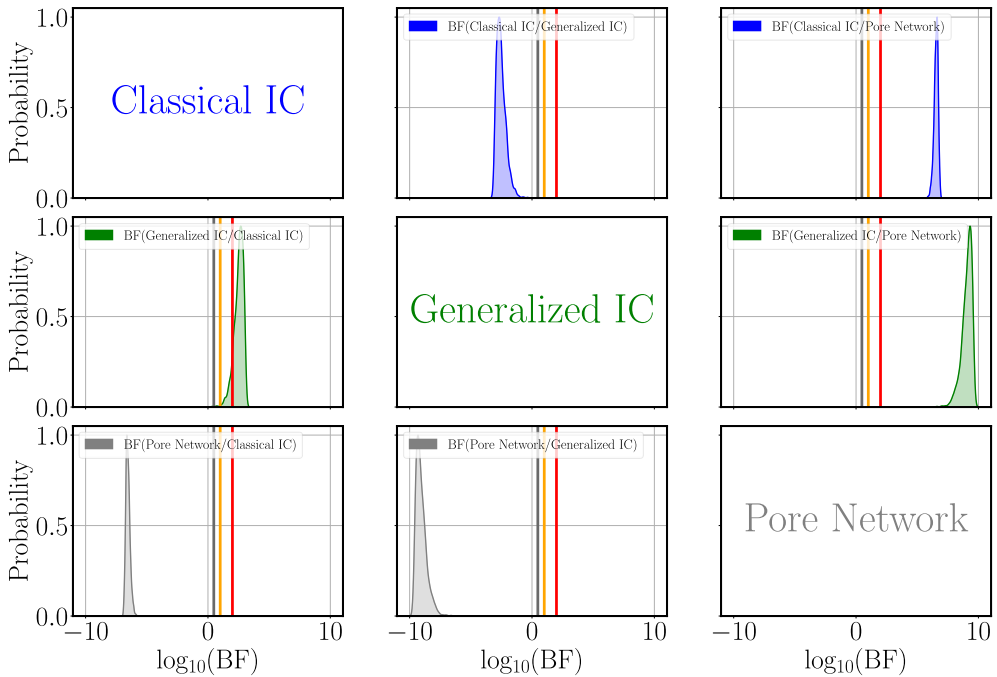


Pressure

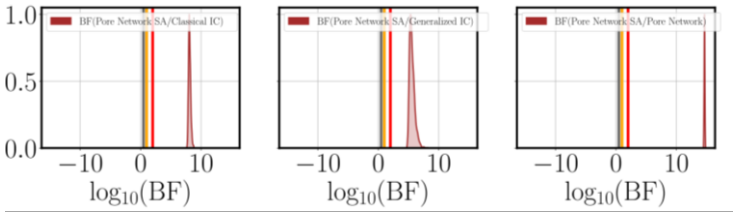


Application: Coupling free flow and porous-medium flow

Models → Reference → Priors → Sensitivity → Posterior → Bayes Factor



“no substantial” evidence
“substantial” evidence
“strong” evidence



Model weights

Classical IC	0.003
Generalized IC	0.997
Pore Network	0.000

Applicability to (SciML) benchmarks

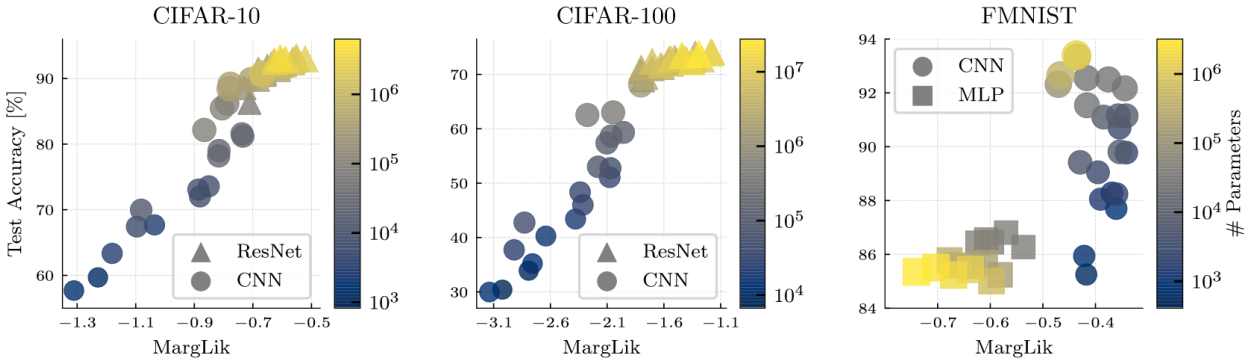
A) Benchmark setup contains uncertainty (uncertain parameters, ...)

→ Direct application possible

B) No inherent uncertainty in the setup

1) Describe selection of model structure using priors

2) Estimate model uncertainty from e.g. numerical properties



Immer. et al. (2021)

Universität Stuttgart

Institute for Modelling Hydraulic and Environmental Systems
Department of Hydromechanics and Modelling of Hydrosystems

Thank you!

Rebecca.Kohlhaas@iws.uni-stuttgart.de
Universität Stuttgart
IWS – LH2

BayesValidRox

SFB 1313

Sources

- Smith (2014) Uncertainty Quantification: Theory, Implementation, and Applications, *SIAM*.
- Kass, R. E., & Raftery, A. E. (1995). Bayes factors. *Journal of the american statistical association*, *90*(430), 773-795.
- Jeffreys, H. (1998). The theory of probability. *OuP Oxford*.
- Schöniger, A. et al. (2014). "Model selection on solid ground: Rigorous comparison of nine ways to evaluate Bayesian model evidence." *Water resources research* 50.12
- Schöniger, A. et al. (2015) "Finding the right balance between groundwater model complexity and experimental effort via Bayesian model selection." *Journal of Hydrology* 531
- Lotfi, S., Izmailov, P., Benton, G., Goldblum, M. & Wilson, A.G.. (2022). Bayesian Model Selection, the Marginal Likelihood, and Generalization. *Proceedings of the 39th International Conference on Machine Learning,* in *Proceedings of Machine Learning Research* 162:14223-14247 Available from https://proceedings.mlr.press/v162/lotfi22a.html
- Mohammadi, F. (2023). *A surrogate-assisted Bayesian framework for uncertainty-aware validation benchmarks*. Stuttgart: Eigenverlag des Instituts für Wasser-und Umweltsystemmodellierung der Universität Stuttgart.
- Radev, S. T., et al. (2021) "Amortized bayesian model comparison with evidential deep learning." *IEEE Transactions on Neural Networks and Learning Systems* 34.8
- Mohammadi, F., et al. (2023) "A surrogate-assisted uncertainty-aware Bayesian validation framework and its application to coupling free flow and porous-medium flow." *Computational Geosciences* 27.4
- Berre, I., et al. (2021) "Verification benchmarks for single-phase flow in three-dimensional fractured porous media." *Advances in Water Resources* 147